Introduction to Sample size calculations

Project: Multivariate and Hierarchical Data

Yannick Vandendijck

Janssen Pharmaceutica, Johnson & Johnson
&
I-Biostat, Hasselt University







Interuniversity Institute for Biostatistics
and statistical Bioinformatics



Why to bother?

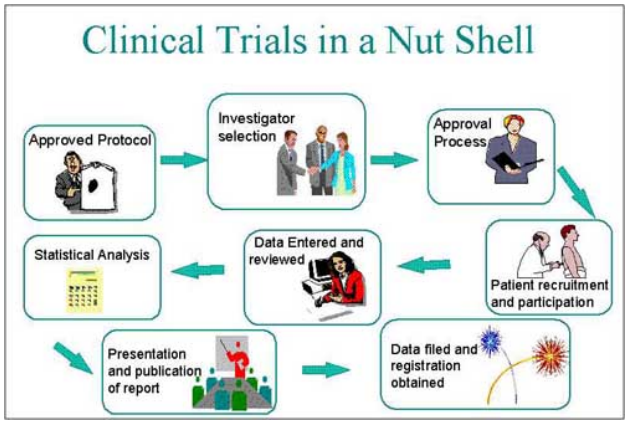
Two frameworks

- **All data available**: Sabermetrics



mindforsports.com

- **Experiments to be done**: Clinical trials



padisysmobileclinicaltrials.wordpress.com

Experiments to be done

- Clinical Trials - **Drug/Vaccine development**:
  - Q: Is the drug/vaccine safe? Is the drug/vaccine working?
  - Randomized controlled trials
  - Carefully defined and operationalized studies that involve a well-defined study design (and thus sample size)

Experimental design

- What is the (primary) research question?
- How to conduct the experiment?
  - What is your primary endpoint?
  - Which is the population of interest? (inclusion/exclusion criteria)
  - Which covariates need to be controlled? (eg, temperature in lab)
  - How to apply randomization?
  - How to obtain measurements?
  - Operational/Technical possibilities?
- **What is the sample size?**

Experimental design

"as ye shall design is as ye shall analyse"

and the reverse

"as ye shall analyse is as ye shall design"

(Steven A. Julious)

Primary endpoint

The primary endpoint and objective should be used for sample size determination!

Choosing the primary endpoint is an essential and vital part of each study:

- "The primary endpoint should be the variable capable of providing the most clinically relevant and convincing evidence directly related to the primary objective of the trial."*
- A good endpoint is well-defined, measurable, valid, reliable, allows for repeatability and minimizes systematic error

* ICH Topic E9: Statistical Principles for Clinical Trials (1998)

Primary endpoint

You want to test whether a new treatment is better than an existing treatment for a specific virus-related disease in hospitalized adults. How

can you measure that the new treatment would be better?:

- Speed of viral shedding
- Length of hospital stay (! very dependent on SoC in a country)
- Vital signs recordings (e.g., oxygen saturation, ...)
- Quality of life measurements
- ...

Think carefully and discuss with scientists/clinicians/Key opinion Leaders... about the primary endpoint!!

Sample size is key in design considerations

- Sample size determines the overall uncertainty of the analysis results
  - $\Rightarrow$ determines precision of parameter estimates
  - $\Rightarrow$ determines power of statistical tests
  - $\Rightarrow$ determines the effect size we will be able to detect

- How many subjects do we need to see a meaningful difference?
- Using available sample size, which difference can we detect that is statistical significant?
- Does it make sense to perform experiment with current budget?

Wrong sample size impacts

Too small sample size?

- We will miss biologically relevant differences as they turn out to be statistically not significant
- Experiment may end up as wasting of time/resources
- We could expose subject to potential harmful treatments without advancing scientific knowledge

Too big sample size?

- We will be able to see effects that are too tiny to be scientifically important
- We will certainly waste resources/harm patients
- Patients are denied a potential beneficial treatment

That is why we need sample size calculation!

Sample size calculations

Points of view:

- **Frequentist approach**
  (length of confidence intervals or **power**)

- Bayesian approach
  (optimizing utility function)

Let's talk about Power

Example: Systolic Blood Pressure

Description of the experiment:

- Two-sample experiment
- Compare systolic blood pressure (SBP) in a treatment group and control group
- SBP measured using a standard sphygmomanometer
- Treatment: we expect SBP to be reduced

Hypothesis testing

- We test H_0 against H_1
- Decision about H_0: rejection or non-rejection
- Decision based on data available at hand

- Rejecting H_0 in favour of H_1 means that there is an **evidence against** H_0 in the data
- Not rejecting H_0 means that there is **lack of evidence against** H_0 in the data (**this does not imply that** H_0 **is true**)

Lack of evidence

What does lack of evidence imply - **2 options**:

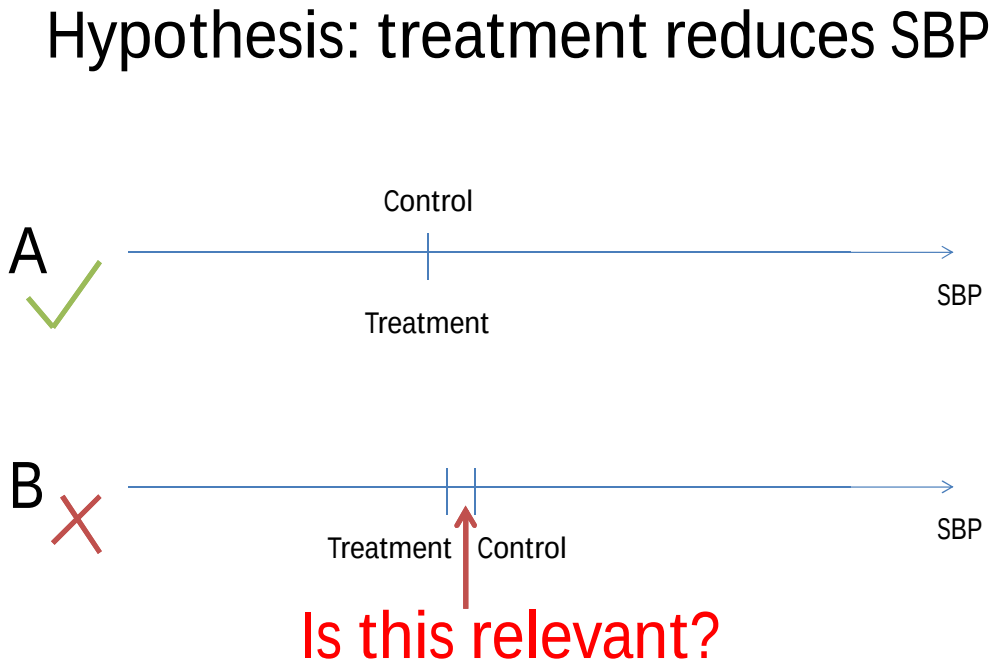
A. The H_0 is true;

B. The H_0 is not true, but we do not have enough **power** to detect it

- **B** translates: the effect size we are able to detect using our data does not turn out to be statistical significant
- Hence, sample size was not adequate and we need more data to detect it

Effect size is "simply a way" of quantifying the size of the difference between two groups
Example: We power a study to detect an effect size of 100, however the true underlying difference is only 30. Our study could fail to reject H_0
$\Rightarrow$ Study was underpowered for the true effect size - we need more data

Lack of evidence options



Always ask yourself/scientists the question whether the effect size assumed is also biologically relevant

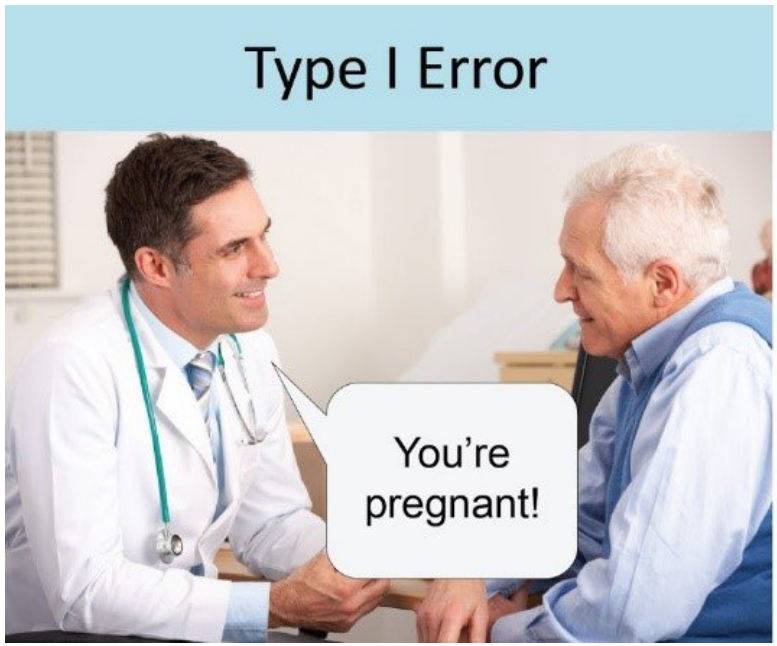
Statistical significant $\neq$ Scientifically relevant

Possible decisions

	H_0 true	H_0 false
Reject H_0	Type I error	
Non-rejection H_0		Type II error

Possible decisions

H_0: Not pregnant vs H_1: Pregnant





Type I and Type II errors

In SBP example:

- **Type I error:**
  When (in truth) there is no treatment effect, the data might suggest a treatment effect

  Type I error $= P(H_1|H_0)$
- **Type II error:**
  When (in truth) there is a treatment effect, the data might suggest no treatment effect

  Type II error $= P(H_0|H_1)$

Trade off between Type I and Type II errors

- Both errors **cannot be minimized simultaneously**
- I will always reject H_0: Type II $= 0$, but Type I $= 1$
- I will never reject H_0: Type II $= 1$, but Type I $= 0$
- Strategy: we control one (Type I error) and then we try to minimize the other (Type II other)

	H_0 true	H_0 false
Reject H_0	Type I error	✓
Non-rejection H_0	✓	Type II error

Type I error

- Probability of false rejection H_0
- $P(H_1|H_0)$
- Translate: falsely finding effect, where there is none
- Typically fixed as **significance level** (*alpha* level)
- Can be translated as the risk of the *regulatory health agencies*

$\Rightarrow$ Type I error is controlled to be under certain threshold α

Type II error

- Probability of missing rejection of H_0
- $P(H_0|H_1)$
- Translate: I do not reject, when there is -however- an effect in reality
- The study did not succeed its objectives
- It is a probability $\Rightarrow$ it depends on true underlying effect size!
- Can be translated as the risk by the *sponsor of the study*
- We usually talk about **power** instead of Type II error

Power

- 1 - Type II error = 1 - $P(H_0|H_1) = P(H_1|H_1)$
- Probability of rejecting H_0 when indeed H_0 is not true
- Translate: we find the true effect
- Simple relationship: We can increase power with increasing sample size
- As Type II error, it depends on true underlying effect size!

Power and Sample size

Power of the experiment depends on

- Test and hypotheses: fixed
- Desired significance level: fixed
- True underlying effect: cannot be influenced
- True underlying variability: cannot be influenced
- **Sample size**: only item that we can fully control

Summary

Why to do sample size calculation?

- We seek for biologically relevant effect size
- **Sample size** influences **power**
- **Power** determines the ability to **rightfully reject** null hypothesis H_0 for biologically relevant effect size
- **Rejection** of H_0 validates the observed effect size as **statistically significant**
- **Statistically significant effect size that is biologically relevant** is what we seek for

Steps in a sample size calculation

Example: Systolic Blood Pressure

Description of the experiment:

- Two-sample experiment
- Compare systolic blood pressure (SBP) in a treatment group and control group
- SBP measured using a standard sphygmomanometer
- Treatment: we expect SBP to be reduced

Question:
How many patients should we include in the treatment and control group?

Steps in a sample size calculation

Elements to determine sample size

1. Specify parameter, hypothesis and test
2. Specify significance level
3. Specify effect size
4. Obtain values or estimates of other parameters needed
5. Specify a target value for the power

Step 1: Specify parameter and hypothesis

- **First and very important step!**
- **What do you want to test**, what is the goal of the study?

- SBP: Treatment is supposed to reduce blood pressure, thus:

H_0 : $\mu_T \geq \mu_C$, i.e. mean SBP same for treatment and control

H_1 : $\mu_T < \mu_C$, i.e. mean SBP smaller for treatment

- Parameter that is tested: $\delta = \mu_T - \mu_C$:

$$H_0 : \quad \delta \geq 0$$
$$H_1 : \quad \delta < 0$$

Step 1: Specify test on parameter

- Two samples: mean comparison
- Let us assume normality and homoscedasticity
- **Test**: two sample t-test with one sided hypothesis

Step 2: Specify significance level

- **Significance level** = threshold on the Type I error probability
- Type I error = $P(H_1|H_0)$
- Usually ranging between 0.001 to 0.1
- Most of the time: $\alpha = 0.05$

Step 2: Specify significance level

"The value for which P=0.05, or 1 in 20, is 1.96 or nearly 2; it is **convenient to take this point as a limit in judging whether a deviation ought to be considered significant or not**. Deviations exceeding twice the standard deviation are thus formally regarded as significant. Using this criterion we should be led to follow up a false indication only once in 20 trials, even if the statistics were the only guide available."

(R.A. Fisher, Statistical Methods for Research Workers, 1925)

Step 2: Specify significance level

"... **Consequently, isolated significant results which he does not know how to reproduce are left in suspense pending further investigation.**

(R.A. Fisher, The Statistical Method in Psychical Research, 1929)

Step 3: Specify effect size

Effect size: the treatment effect you expect/anticipate to detect with your study

To define effect size, ask yourself/scientist questions like:

- "What would be the minimal improvement for patients taking the new medication?"
- "What results do you expect?"
- "What treatment effect is of scientific interest?"

Step 3: Specify effect size

SBP example:

- Relevant treatment effect might be -15 mm Hg
- Translate: compared to control, the treatment reduces blood pressure with 15 mm Hg
- If you were the patient, the benefits of reducing SBP by 15 mm Hg outweigh the cost, inconvenience, and potential side effects of this treatment

Specifying an effect size of 15 mm Hg means that in case we run our study and we observe a difference between the control and treatment group of around 15 or more, it will turn out to be statistically significant (given that the sample size calculation was correct!)

Step 3: Specify effect size

BUT keep in mind that:

- The smaller the effect size, the larger the sample size
- Be reasonable with choice of effect size
- Should be determined together with scientist/clinical team
- Problem need to be clearly communicated

Step 3: Specify effect size

Example of issues:

MD Go ahead, reduce sample size, because we are pretty sure that treatment will do better than this!

St But what if not? What is the clinically relevant effect size?

MD [This is the effect we expect to see. That is our effect size].

MD I think power 70% is really enough for us. I see we can reduce the number of patients by decreasing the power.

St But what is the relevance of such a result? Is 70% worth even conducting experiment?

MD This book says that medium effect size is five.

St Effect size is always context dependent, we are not interested in experiment of anyone else, we need an effect size relevant for you.

Another example: https://www.youtube.com/watch?v=Hz1fyhVOjr4

Step 3: Specify effect size

Use of **historical data and previous scientific experience** to adjust effect size:

YES : If overwhelming historical/scientific evidence suggests that your experiment should produce effect sizes in the magnitude of e.g. 20-50, you may specify you effect size of interest to be e.g. 10-15, even if actual minimal scientifically relevant effect size is lower

NO One previous study has shown effect size of e.g. 20, so you specify your effect size as 20 regardless the minimal scientifically relevant effect. There is very little guarantee that your experiment will produce same effect size as previous one. Especially if pilot study is considered as source of effect size estimate.

Be aware of publication bias...!?

Step 3: Specify effect size

Summary:

- Effect size quantifies the expected/anticipated/minimum difference between the groups (effect of the treatment)
- Should be determined based on scientific knowledge
- **Avoid using pilot study to obtain estimate for effect size!**

Step 4: Obtain values or estimates of other parameters

- Power can depend on **other parameters**
- These parameters can be completely unrelated to the hypotheses
- BUT they can highly influence results
- SBP example: variance of the measurements in the planned experiment
- How to find a value for these parameters?
  1. Based on scientific knowledge (experience)
  2. Use historical data
  3. Conduct pilot study

Step 4: Obtain values or estimates of other parameters

1. Based on expert scientific knowledge:
   - Reliability of expert?
   - Scientific consensus about the problem?
   - Novelty of experiment?
   - Do not overestimate nor underestimate expert opinion!

Step 4: Obtain values or estimates of other parameters

2. Use of historical data:

- In-house data
- Data from literature and public databases
- Compare designs
- Reliability of the data?
- Be careful to check that the same parameter is estimated!

Step 4: Obtain values or estimates of other parameters

Issues with historical data/scientific knowledge:

- Manufacturer of sphygmomanometers published: $sd = 2.5\mathrm{mmHg}$, but this reflects variation in readings made on same subject
- In previous study, only women were included, but in planned study mix of gender $\Rightarrow$ variation due to gender not included
- Historical data were collected using different measurement device or even different units

Step 4: Obtain values or estimates of other parameters

3. Conduct pilot study:

- Same experimental design with small sample size
- To estimate additional parameters (typically variance)
- It has to be sufficiently large to be reliable!
- Do not overestimate small pilot studies
- **Should not be used to determine effect size to be studied**

Step 5: Specify a value for the power

How to choose the **power**?

- The power is the probability with which the desired effect size will be detected $= P(H_1|H_1)$
- 80% is most common but also somewhat minimal
- Sometimes higher power is more appropriate (85% or 90%)
- As power increases, required sample size increases fast

Summary for SBP

Choices for SBP example:

1. Specify hypothesis test on parameter:

$$H_0: \quad \delta \geq 0$$
$$H_1: \quad \delta < 0$$

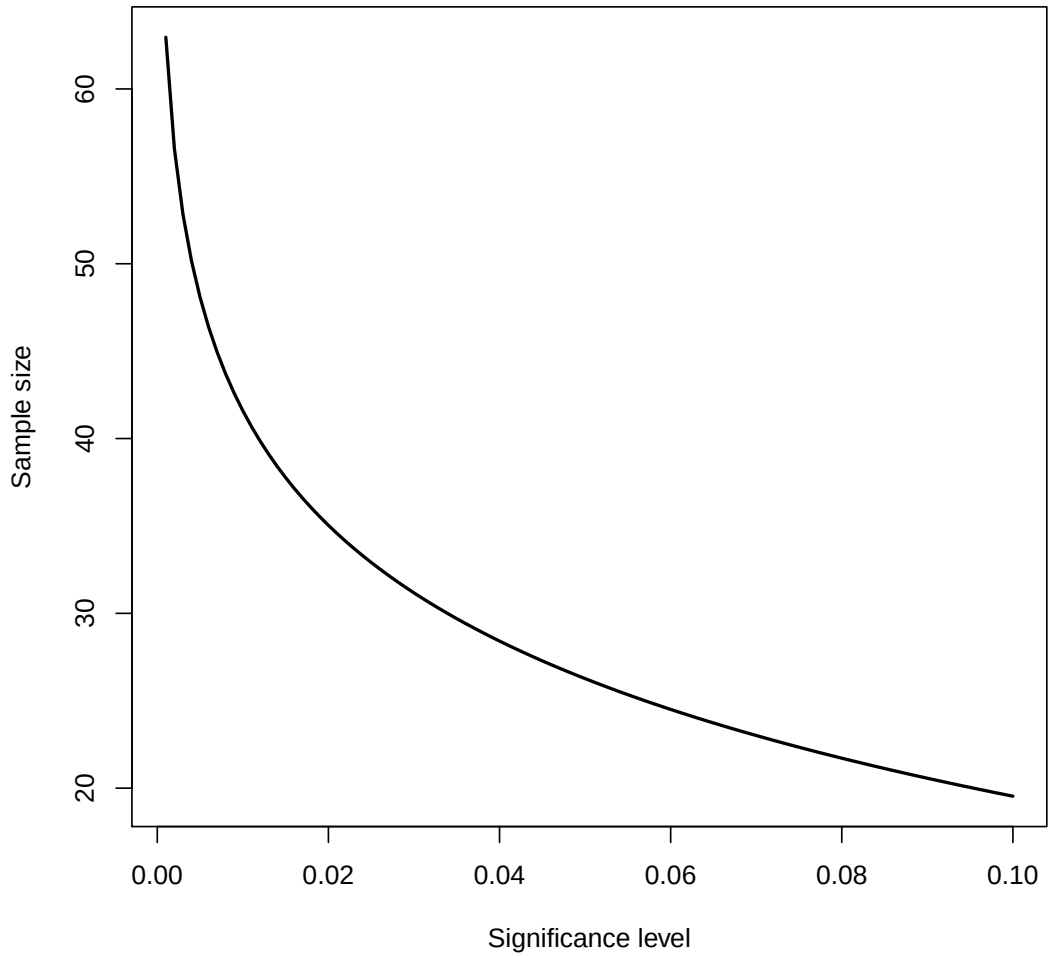
   Tested with one sided t-test.
2. Specify significance level: $\alpha = 0.05$
3. Specify effect size: $\delta = -15\text{mmHg}$
4. Obtain values or estimates of other parameters needed: $sd = 20\text{mmHg}$
5. Specify a value for the power: $\beta = 0.85$

Resulting sample size: 27 patients/group.

Visualization

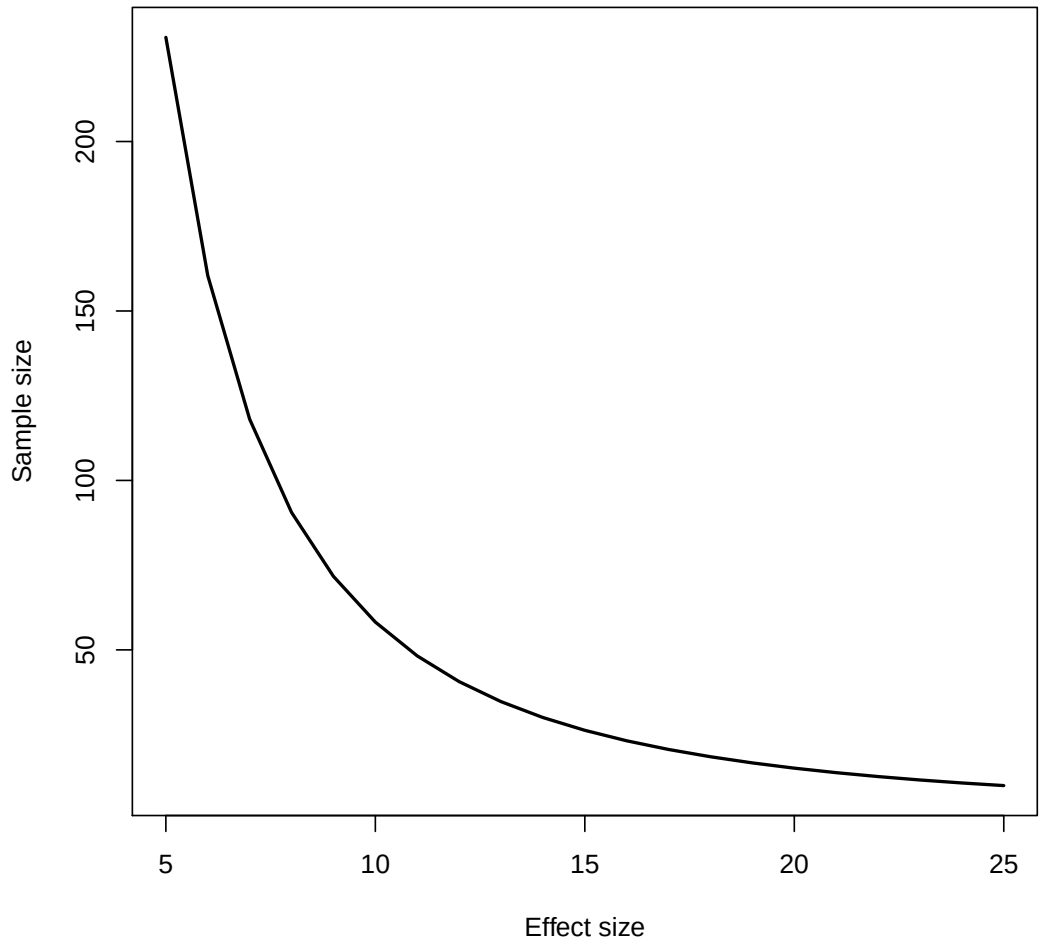
Sample size curves

SBP example: $\alpha =$?, $\delta = 15$, $sd = 20$, $\beta = 0.85$



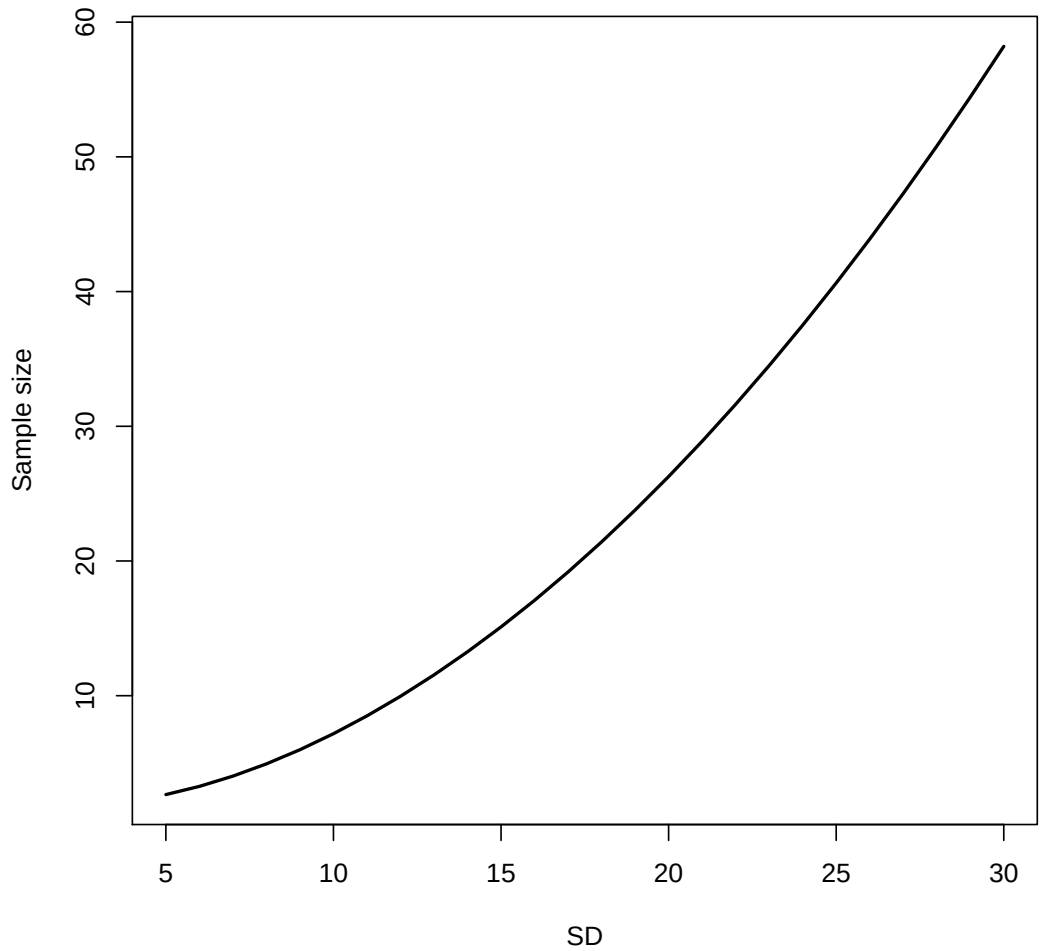
Sample size curves

SBP example: $\alpha = 0.05$, $\delta =$?, $sd = 20$, $\beta = 0.85$



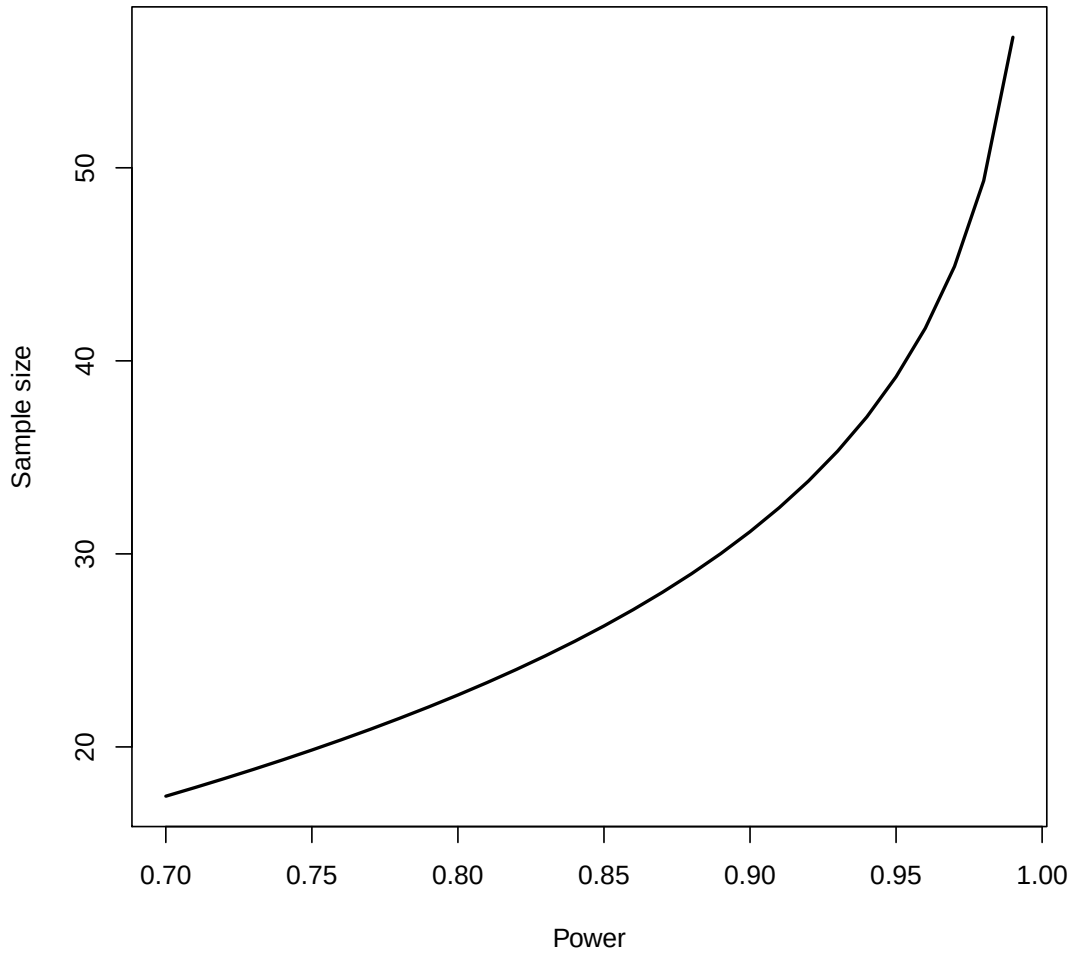
Sample size curves

SBP example: $\alpha = 0.05$, $\delta = -15$, $sd =$?, $\beta = 0.85$



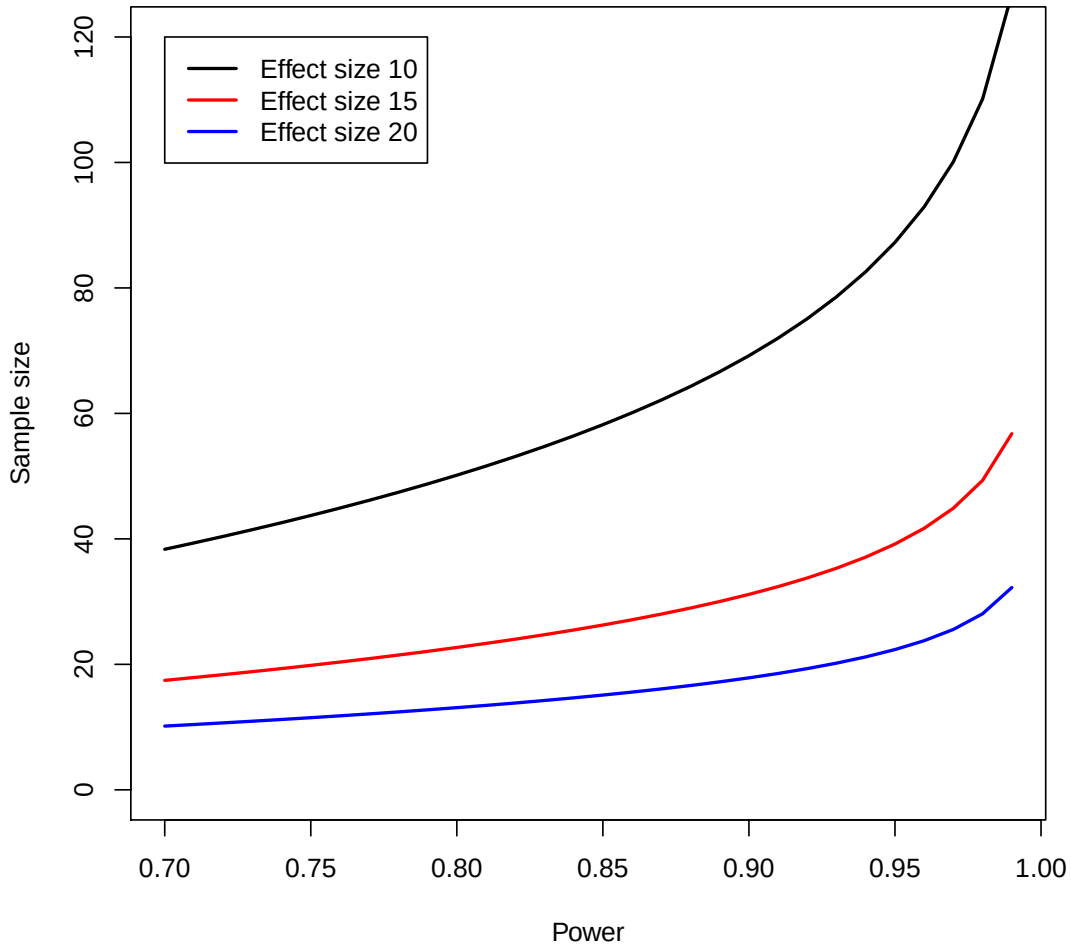
Sample size curves

SBP example: $\alpha = 0.05$, $\delta = -15$, $sd = 20$, $\beta =$?



How it is often presented - Sample size curves

SBP example: $\alpha = 0.05$, $\delta =$?, $sd = 20$, $\beta =$?



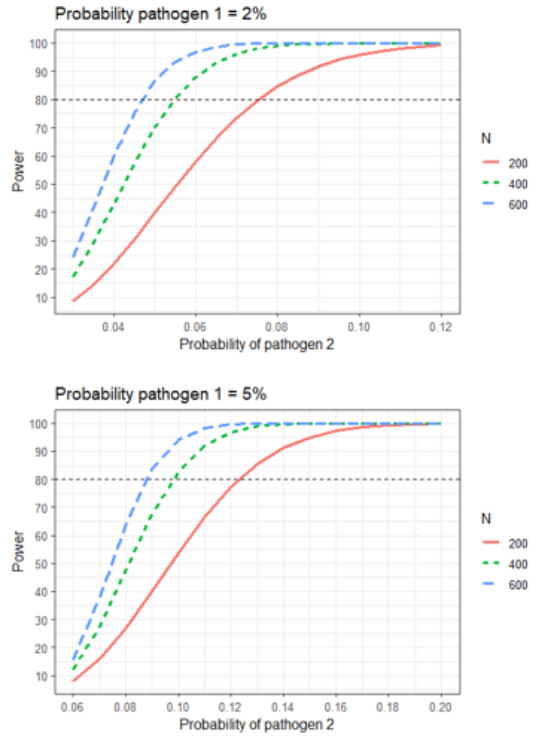
How it is often presented - Sample size tables

δ	β	α	sd	n
10	0.80	0.05	20	51
10	0.85	0.05	20	59
10	0.90	0.05	20	70
15	0.80	0.05	20	23
15	0.85	0.05	20	27
15	0.90	0.05	20	32
25	0.80	0.05	20	9
25	0.85	0.05	20	10
25	0.90	0.05	20	12

How it is often presented - Example

$P_{pathogen\ 1}$	$P_{pathogen\ 2}$	N=200		N=400		N=600	
		Expected cases	Power	Expected cases	Power	Expected cases	Power
2 %	4 %	4 – 8	22 %	8 - 16	43 %	12 – 24	60 %
	6 %	4 – 12	58 %	8 – 24	88 %	12 – 36	97 %
	8 %	4 – 16	85 %	8 – 32	99 %	12 – 48	100 %
5 %	6 %	10 – 12	8 %	20 – 24	12 %	30 – 36	16 %
	8 %	10 – 16	27 %	20 – 32	48 %	30 – 48	64 %
	10 %	10 – 20	54 %	20 – 40	83 %	30 – 60	94 %
10 %	11 %	20 – 22	7 %	40 – 44	10 %	60 – 66	12 %
	13 %	20 – 26	20 %	40 – 52	33 %	60 – 78	46 %
	15 %	20 – 30	39 %	40 – 60	65 %	60 - 90	81 %

Assumptions:
- McNemar's exact test since we have matched pairs data (pathogen information on same subjects)
- We assume independent samples for the power calculation. The actual power can be increased or decreased depending on the associations between the pathogens. Namely, a study design with matched pairs can improve precision of statistical inference in case a positive association is observed.
- One sided test with significance level 5%
- No multiple correction assumed in case we will test multiple pathogen incidences with each other in the study
  - For example, comparing 5 pathogens with each other would give 15 statistictal tests which will inflate the type 1 error



10

Short exercise on Blackboard

Recommended reading:
Russell V. Lenth (2001). Some practical guidelines for effective sample size determination. The American Statistician, 55:3, 187-193.

Calculating a sample size

Theoretical derivation: normal case

This is taught a lot, but only rarely used...

- Two samples comparison
- SBP example
- Simplification: assume that σ is known
- $Y_{Cj} \sim N(\mu_C, \sigma^2)$ values of patients in control group
- $Y_{Tj} \sim N(\mu_T, \sigma^2)$ values of patients in treated group
- $\delta = \mu_T - \mu_C$
- $N =$ **?** sample size per group

Theoretical derivation: normal case

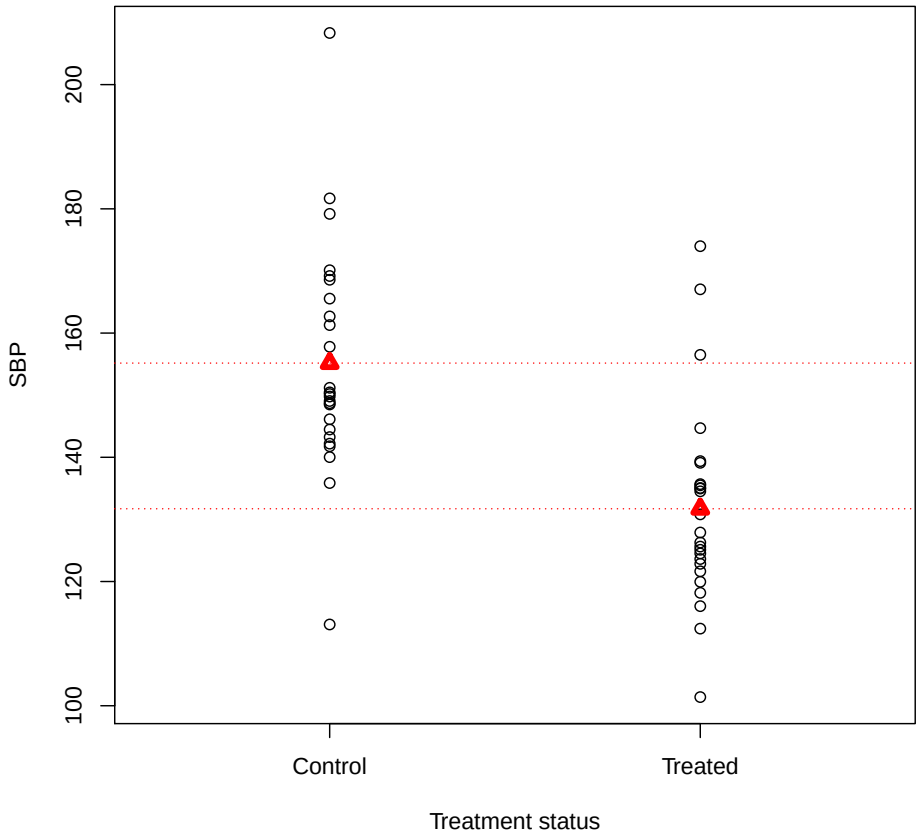
- Test of hypothesis

$$H_0 : \delta \geq 0$$
$$H_1 : \delta < 0$$

- $\bar{Y}_C = \frac{1}{N} \sum_{j=1}^{N} Y_{Cj}$
- $\bar{Y}_T = \frac{1}{N} \sum_{j=1}^{N} Y_{Tj}$
- $\hat{\delta} = \bar{Y}_T - \bar{Y}_C$

- That implies

$$\bar{Y}_T - \bar{Y}_C \sim N\left(\mu_T - \mu_C, \frac{2\sigma^2}{N}\right)$$
$$\hat{\delta} \sim N\left(\delta, \frac{2\sigma^2}{N}\right)$$



Theoretical derivation: normal case

- We assume that σ is known, thus we can use Z-test
- Therefore, from $\hat{\delta} \sim N\left(\delta, \frac{2\sigma^2}{N}\right)$ it holds $\frac{\hat{\delta} - \delta}{\sqrt{2\sigma^2}} \cdot \sqrt{N} \sim N(0,1)$
- Under H_0: $\frac{\hat{\delta}}{\sqrt{2\sigma^2}} \cdot \sqrt{N} \sim N(0,1)$
- Hence $Z = \frac{\hat{\delta}}{\sqrt{2\sigma^2}} \cdot \sqrt{N}$ is our test-statistic
- Evidence against $H_0: \ \delta \geq 0 \Rightarrow$ if Z is high negative number

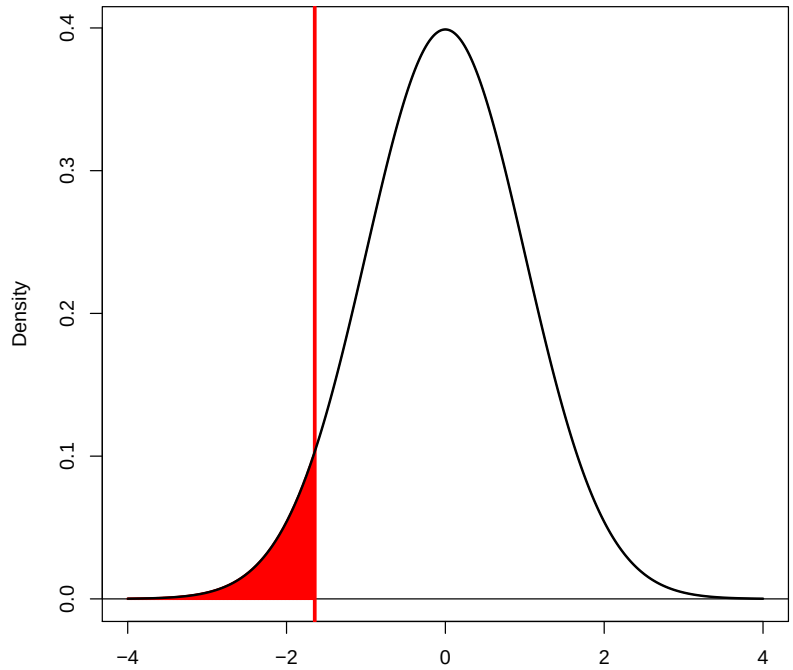
Theoretical derivation: normal case

Under null hypothesis:

- $P_{H_0}(Z < z_\alpha) = \alpha$

$\Rightarrow$ rejecting if $Z < z_{0.05}$ (we have only 5% chance of wrong rejection under H_0)

$\Rightarrow$ rejecting if $\hat{\delta} < z_\alpha \cdot \sqrt{\frac{2\sigma^2}{N}}$

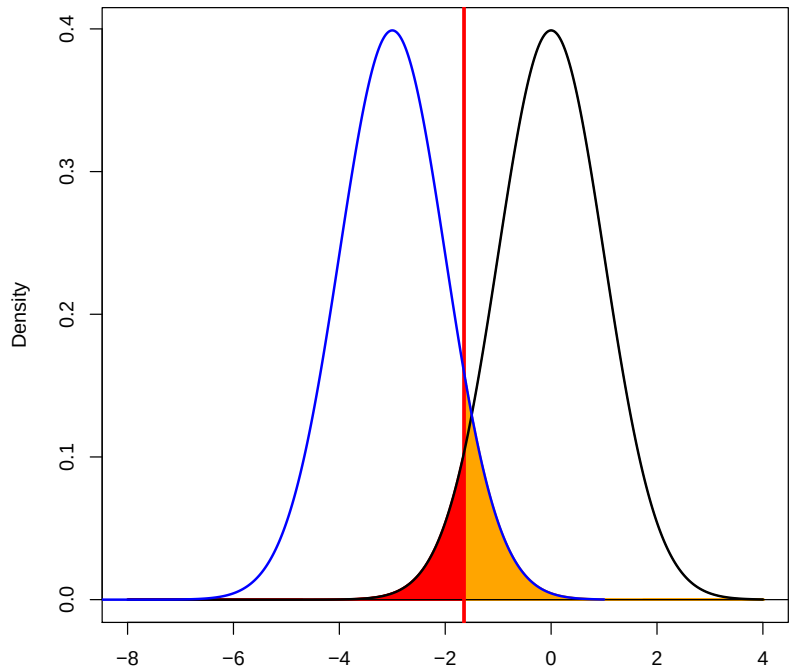


Theoretical derivation: normal case

Include the power in the calculations

Under alternative hypothesis:

- $\frac{\hat{\delta} - \delta}{\sqrt{2\sigma^2}} \cdot \sqrt{N} \sim N(0, 1)$
- and we want
  $P_{H_1}\left(\hat{\delta} < z_\alpha \cdot \sqrt{\frac{2\sigma^2}{N}}\right) = \beta$
- subtract δ and multiply by $\sqrt{\frac{N}{2\sigma^2}}$ to get variable with $N(0, 1)$
- $P_{H_\delta}\left(\frac{\hat{\delta}-\delta}{\sqrt{2\sigma^2}}\sqrt{N} < z_\alpha - \delta\sqrt{\frac{N}{2\sigma^2}}\right) = \beta$



Theoretical derivation: normal case

Formula for sample size

- $\Phi\left(z_\alpha - \delta\sqrt{\frac{N}{2\sigma^2}}\right) = \beta$
- $z_\alpha - \delta\sqrt{\frac{N}{2\sigma^2}} = z_\beta$
- Finally, formula for samples size calculation:

$$N = 2 \cdot (z_\alpha - z_\beta)^2 \cdot \frac{\sigma^2}{\delta^2}$$

Theoretical derivation: normal case

Apply the formula on the SBP example:

$$N = 2 \cdot (z_\alpha - z_\beta)^2 \cdot \frac{\sigma^2}{\delta^2}$$

- $\alpha = 0.05$, thus $z_\alpha = z_{0.05} = -1.645$
- $\beta = 0.85$, thus $z_\beta = z_{0.85} = 1.036$
- $\sigma = 20$ (!! it is assumed known here)
- $\delta = -15$

Plugging all the numbers in gives N=26 (cfr. with 27 obtained previously)

Theoretical derivation: normal case

Notation comment:

- Our final formula:

$$N = 2 \cdot (z_\alpha - z_\beta)^2 \cdot \frac{\sigma^2}{\delta^2}$$

- Note: $z_\alpha = -z_{1-\alpha}$

$$N = 2 \cdot (z_{1-\alpha} + z_\beta)^2 \cdot \frac{\sigma^2}{\delta^2}$$

- We denote β as power, but sometimes $1 - \beta$ denotes power. Then:

$$N = 2 \cdot (z_\alpha - z_{1-\beta})^2 \cdot \frac{\sigma^2}{\delta^2}$$

- BE CAREFUL WHEN USING LITERATURE!

Theoretical derivation: normal case - Extensions of the framework

Theoretical derivation: normal case

Two-sided hypothesis

- Theoretical derivation assumed one sided hypothesis
- What if hypothesis would be two-sided:

$$H_0: \quad \delta = 0$$
$$H_1: \quad \delta \neq 0$$

- Then for normal case:

$$N = 2 \cdot (z_{\frac{\alpha}{2}} - z_\beta)^2 \cdot \frac{\sigma^2}{\delta^2}$$

Theoretical derivation: normal case

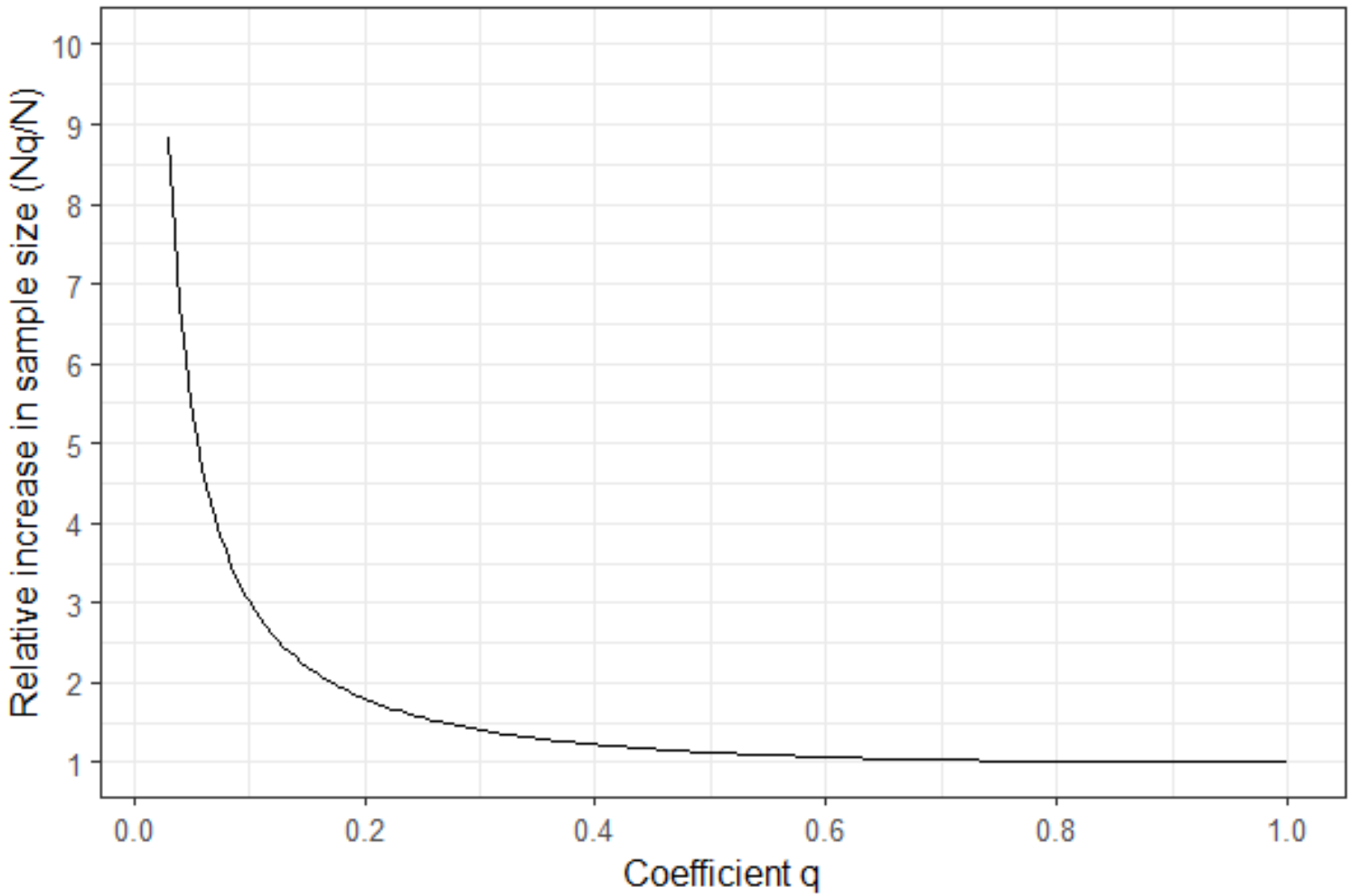
Unbalanced design

- Theoretical derivation assumed same number of patients in both groups
- It can occur that we want to plan an unbalanced design
- One group has N observations, other one qN, with $q \in (0, 1]$
- Then for normal case:

$$N = \frac{q+1}{q} \cdot (z_\alpha - z_\beta)^2 \cdot \frac{\sigma^2}{\delta^2}$$

Theoretical derivation: normal case

Unbalanced design: increment in sample size



Theoretical derivation: normal case

Heteroscedasticity

- Theoretical derivation assumed same variance for both groups
- What if variance is different, σ_1^2 and σ_2^2?
- The variances are still assumed known
- Then for normal case:

$$N = \frac{\sigma_1^2 + \sigma_2^2}{\delta^2} \cdot (z_\alpha - z_\beta)^2$$

Calculating a sample size - More real-life applications

Case of t-test

- Variance σ^2 unknown
- In final experiment, variance will be estimated from the data
- If we adjust for this, we need to use t-distribution quantiles $t_{\alpha,n}$
- **Degrees of freedom n depends on sample size**
- Iterative procedures are necessary
- **Software solutions** quite straightforward
- By hand: normal case as approximation (big samples)

Comparison of more groups

- What if we have more groups to compare?
- One-way ANOVA setting
- Getting difficult, because degrees of freedom of F distributions depends on N
- Iterative procedures are necessary
- **Software solutions** quite straightforward

Comparison of more groups - ANOVA mean structure

- Example: effect size of interest $\delta = 5$; assume 5 groups
- Nuisance parameter: *within variability* (σ_e^2)
- *Between variability*: depends on mean structure!
  - Option 1: 10 - 12.5 - 12.5 - 12.5 - 15 $\Rightarrow$ $BW = 3.125$
  - Option 2: 10 - 10 - 10 - 10 - 15 $\Rightarrow$ $BW = 5.00$
  - Option 3: 10 - 10 - 10 - 15 - 15 $\Rightarrow$ $BW = 7.50$
- Different between variabilities $\Rightarrow$ which one to choose?! $\Rightarrow$ smaller BW will give larger sample size
- Possible solution: determine sample size for all of them and retain the one which yields highest sample size

Binary response

- Response of interest has binary value 0/1
- Parameter π represents probability of success - **The variance also depends on** π
- Two-groups comparison setting
- Control $X \sim Be(\pi_C)$ and Treatment $Y \sim Be(\pi_T)$
- We are interested in difference between success probabilities $\delta = |\pi_T - \pi_C|$ (effect size)

Binary response

- Then one sided hypothesis yields sample size formula:

$$N = (z_\alpha - z_\beta)^2 \cdot \frac{\pi_T(1-\pi_T) + \pi_C(1-\pi_C)}{\delta^2}$$

- Problem: unknown π_T and π_C appears in formula

- You need to have an idea about their value, not only their difference δ

Poisson response

- Count data
- Represents number of events during some period of time
- No restriction on maximum
- Parameter λ represents frequency of events
- If $\lambda > 20$ asymptotic starts to work: $X \sim Pois(\lambda) \Rightarrow X \approx N(\lambda, \lambda)$
- Then, comparing two frequencies is same as comparing two means of normal distributions, while variances are different
- Otherwise, more elaborate **software solutions** are needed

Multiple comparisons

- Several tests are performed simultaneously
- We want to be protected against any false finding (family-wise error rate)
- We can have multiple t-test, but also combination of the tests
- How to incorporate this into the sample size calculation framework?
- Adjusting confidence level is usually sufficient solution (e.g. Bonferroni)
- **Simulations** can provide solution in more complex cases

Simulation: The universal solution

Simulation method

- Possible in any setting
- Especially useful in complex situations

1. Fix sample size at N
2. Fix other parameters: δ, α
3. Simulate huge number of experiments and test
4. Estimate power
5. Change N accordingly and run 1-4 until you reach desired power
6. Use final N as your sample size

Simulation method: (1) initial N

- Use "good guess"
- It does not really matter
- Only purpose is to get idea of power for this N
- Then change N accordingly
- Possible to run over grid of N

Simulation method: (2) other parameters

- This step is essentially same as in traditional sample size calculation
- You need to fix effect size, variance, α etc.

Simulation method: (3) simulate the experiment

- Core step and often the trickiest to code correctly
- Simulate experiment that has all the properties as you have fixed
- E.g. two-sample t-test: simulate two groups with difference δ in means, SD as fixed in previous step and with N fixed
- It may get more laborious for more complex situations

- Perform the appropriate statistical test and retain p-value
- Repeat this step B times, with sufficient B
- Can be computationally intensive

Simulation method: (4) get power and repeat

- Compare obtained p-values with α level
- How many tests are significant? (eg, $p_i < \alpha$, for i=1,...,B)
- Proportion of significant test among B replicates equals power
- If resulting power is too high or too low, change initial N and repeat the exercise

Simulation method: examples

- Examples follow in provided R codes
- Possible in SAS and other software as well
- Be aware of its computational intensity

Software

General

- Formulas for complicated cases are complex or don't exist
- Once they are programmed, their application is usually straightforward
- Be very careful with any software (read documentation)
- Even commercial software can be misleading

Free software

- R
- Online tools
- Java applications (Russel Lenth)
- G*Power

Commercial software

- SAS, Stata, Statistica, StatXact, SPSS,...
- These statistical software packages have procedures to calculate sample size
- Sometimes dealing only with simple cases

Specialized software often handles more complex situations:

- nQuery, N
- PASS
- East (Cytel)
- Pharsight Trial Simulator

Additional comments

Missing observations

- Computed sample size applies for complete data set
- Actual power can be smaller when observations are missing (eg, patients drop out)
- Necessary to adjust our sample size calculations to the missingness
- Simply by adjusting resulting N with constant representing amount of dropouts
  - If N=250 and we expect 15% drop-out, than
    $N_{adj} = N/(1 - 0.15) \approx 295$
    NOT 250×1.15
- Pilot study or literature can give us an idea on drop-out rate

Adaptive designs

Outside of the scope of this course

- Interim Analysis
- Assessments of futility
- Assessments of efficacy
- Sample size re-estimation (blinded or unblinded)
- Group sequential designs

Russ Lenth's advices

NOT to do

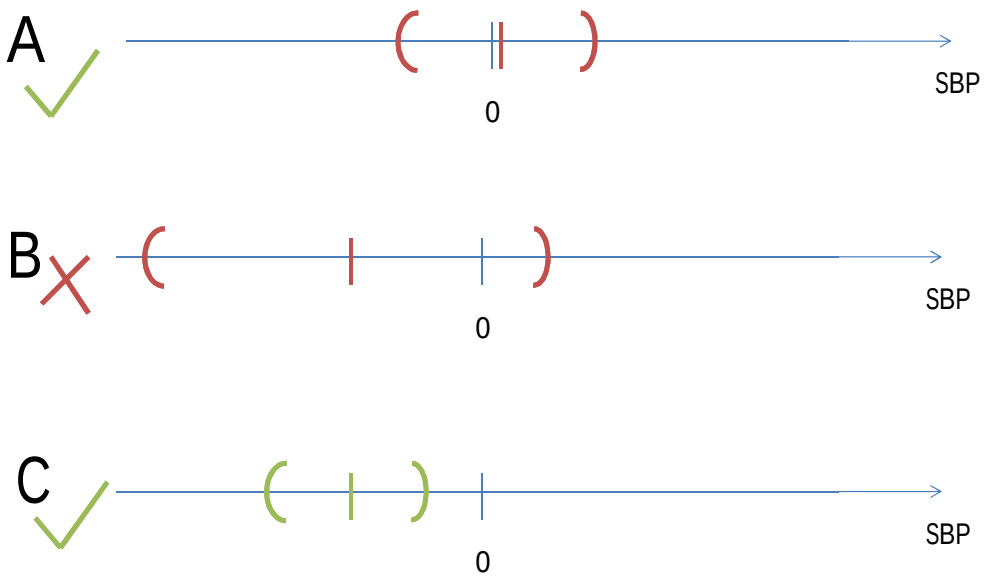
- Retrospective power: It says same thing as p-value, there is no extra information
- Specify T-shirt effect sizes

GOOD to do

- Use power prospectively
- Put science before statistics
- Do pilot studies

Confidence intervals

- Give us additional information about the precision
- They actually tell us most likely value of true effect
- Correct thing to look at instead of "observed power"



Sample size for confidence intervals

- Base sample size on length d instead of effect δ
- Power: What is the smallest effect we want to see significant?
- CI: What is the precision of estimation we want to achieve?
- Similar theoretical background
- Different interpretation and outcome

References

- Lenth, R. V. (2001). "Some Practical Guidelines for Effective Sample Size Determination," *The American Statistician*, 55, 187-193.
- Hoenig, John M. and Heisey, Dennis M. (2001). "The Abuse of Power: The Pervasive Fallacy of Power Calculations for Data Analysis," *The American Statistician*, 55, 19-24.
- Machin, David and Campbell, Michael Joseph (1987). *Statistical tables for the design of clinical trials*, Oxford
- Faes, Christel, Molenberghs, Geert, Aerts, Marc, Verbeke, Geert and Kenward, Michael G. (2009). "The Effective Sample Size and an Alternative Small-Sample Degrees-of-Freedom Method," *The American Statistician*, 63

Acknowledgements for materials

- Thomasz Burzykowski
- Luc Bijnens
- An Cremers
- Helena Geys
- Martin Otava
- Ziv Shkedy